Agile Project Planning with Jira — Sprint Reflection

Lab 2 - Software Engineering

Problem Statement #38 - Dropshipping Inventory & Order Orchestrator

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1. Backlog Summary

The five functional requirements and two non-functional requirements from Lab 1 were decomposed into 4 epics and 15 user stories, all written in the As-a / I-want / So-that form, prioritised High / Medium / Low by user impact, and estimated on the Fibonacci scale.

Epic	Stories	Points	Requirements	Priority mix
Storefront Order Ingestion & Webhook Reliability	4	16	FR-001, NFR-001, NFR-002	3 High, 1 Med
Supplier Integration & Real-Time Inventory Verification	3	18	FR-003, FR-001	2 High, 1 Med
Multi-Vendor Order Splitting & PO Dispatch	4	24	FR-001, FR-002	2 High, 2 Med
Shipment Tracking & Order Visibility	4	21	FR-004, FR-005	2 High, 1 Med, 1 Low
Total	15	79	—	9 High, 5 Med, 1 Low

Sprint	Committed	Completed	Carried over	Velocity
Sprint 1 (1 week)	29 pts / 6 stories	26 pts / 5 stories	3 pts / 1 story	26
Sprint 2 (1 week, in flight)	27 pts / 6 stories	8 pts done, 11 in progress	—	—

2. Did your estimations reflect the actual effort?

Mostly, but not uniformly. The estimates held well at the two ends of the scale and broke down in the middle.

- The 3-point stories that were genuinely mechanical — acknowledging webhooks inside the 200 ms budget (DIOO-6) and capturing supplier tracking numbers (DIOO-16) — finished close to their estimate. Small, single-surface work was estimated reliably.
- The 8-point stories were correctly identified as the hardest items. Real-time supplier stock verification (DIOO-10) and multi-vendor order splitting (DIOO-12) both touch external APIs, failure handling and order state, and both consumed a disproportionate share of their sprint. An 8 meant “this will dominate the sprint”, and it did.
- The clearest miss was DIOO-8, “Quarantine malformed or unsigned webhooks”, estimated at 3. It was sized as if it were only a validation check, but it actually carries signature verification, a dead-letter queue and an alerting path — three integration points. It was the one story that did not finish in Sprint 1 and had to be carried into Sprint 2. In hindsight it was a 5.

The pattern is that effort was underestimated wherever a story silently implied a new piece of infrastructure rather than new logic. Committing 29 points against a delivered velocity of 26 also shows the plan was about 10% optimistic overall — a small, healthy error, but consistently in the same direction.

3. Was your backlog well-prioritised?

Largely yes. Priority was assigned by user impact rather than by technical convenience, and the ordering follows the critical path of the product: nothing in this system has value until an order can be ingested, so the ingestion epic sits at the top; visibility and reporting sit at the bottom.

- All nine High-priority stories map directly to FR-001, FR-002 and FR-004 — the requirements that make the orchestrator functional at all. Sprint 1 pulled the four highest-impact ingestion and inventory stories first, which is the correct order.
- Supplier performance metrics (DIOO-19) was correctly marked Low. It is a reporting convenience, it depends on several sprints of fulfilment history existing first, and it is still in the backlog.
- The weakness: DIOO-8, a Medium, was pulled into Sprint 1 as filler to reach a round commitment, and it became the one story that slipped. A stricter rule — finish every High in an epic before pulling a Medium from it — would have produced a cleaner sprint.

A second refinement worth making is that NFR-001 and NFR-002 are represented only indirectly, through stories inside the ingestion epic. That keeps them visible on the board, but it also means a non-functional requirement can be quietly deprioritised whenever its host epic is deprioritised.

4. How did your simulated sprint align with your plan?

Sprint 1 aligned closely: 26 of 29 committed points were delivered, a 90% completion rate, and the sprint goal — reliable webhook ingestion plus live supplier stock verification — was met. The single 3-point shortfall did not block the sprint goal, which is the outcome a good commitment should produce.

Sprint 2 shows the cost of that overrun. It opened carrying DIOO-8 and committed 27 points, marginally above the observed velocity of 26. At the point of reporting, two stories are Done (8 points), two are In Progress (11 points) and two remain in To Do (8 points), so roughly a third of the sprint is banked and the sprint is tracking to a similar 85–90% outcome rather than a clean 100%.

The structural lesson is that a carryover is not free capacity. Sprint 2 effectively planned 24 points of new work plus 3 points of debt, and the debt arrived already half-finished and half-forgotten. Planning the next sprint at 24–26 points of new work, with carryover counted against the commitment rather than added on top, would align plan and outcome more tightly.

5. What did the burndown chart reveal about capacity?

- Actual capacity is 26 points per sprint, not 29. The remaining-work line never reaches zero — it flattens at 3 and stays there to the sprint boundary. That residual is the honest measure of the team's ceiling, and it should become the input to the next commitment.
- Work did not burn down evenly against the guideline. The actual line tracks above the ideal line for most of the sprint and then drops sharply, which is the classic signature of stories being finished in a batch near the end rather than flowing continuously. In a real team this is an integration-risk signal: if anything had gone wrong late, several stories would have failed together instead of one.
- Scope was stable. No stories were added mid-sprint in Sprint 1, so the entire gap between the guideline and the actual line is execution, not scope creep — which is what makes the velocity figure trustworthy.
- Forecast: with 26 points of backlog still unstarted and 19 points in flight in Sprint 2, at a velocity of 26 the remaining scope is roughly two more sprints. That projection is only usable because the burndown gave a measured velocity instead of an assumed one.

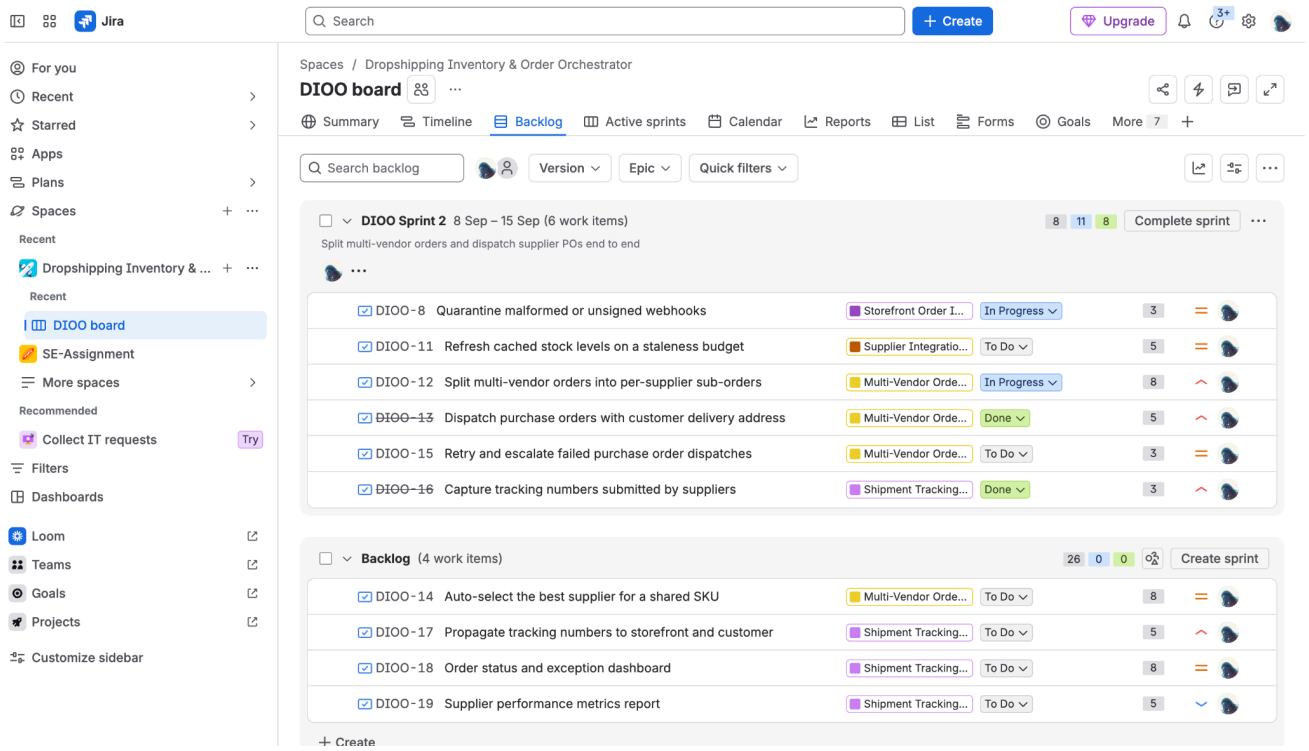
The broader insight is that the burndown is more useful as a diagnostic of flow than as a progress bar. The completion percentage alone said Sprint 1 went well; the shape of the curve said the work was being finished too late in the sprint and that the commitment was set slightly above what the team could actually absorb.

6. Actions for the next sprint

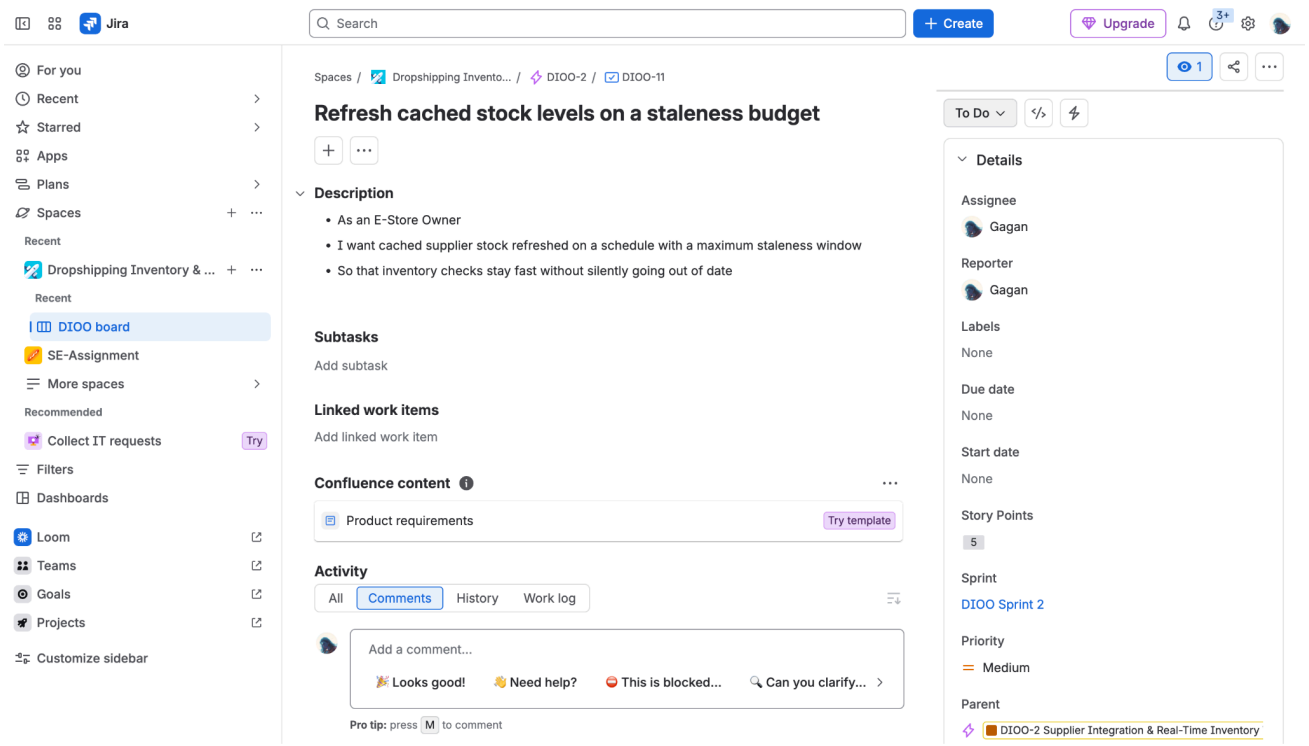
- Cap the Sprint 3 commitment at 26 points, counting any carryover inside that number.
- Re-estimate DIOO-8 from 3 to 5 and split any story that implies new infrastructure rather than new logic.
- Finish every High-priority story in an epic before pulling a Medium from it.
- Give NFR-001 and NFR-002 explicit stories of their own so that non-functional work cannot be deprioritised by accident.
- Add a mid-sprint checkpoint so the burndown is inspected while the sprint can still be corrected, not after it closes.

Screenshots:

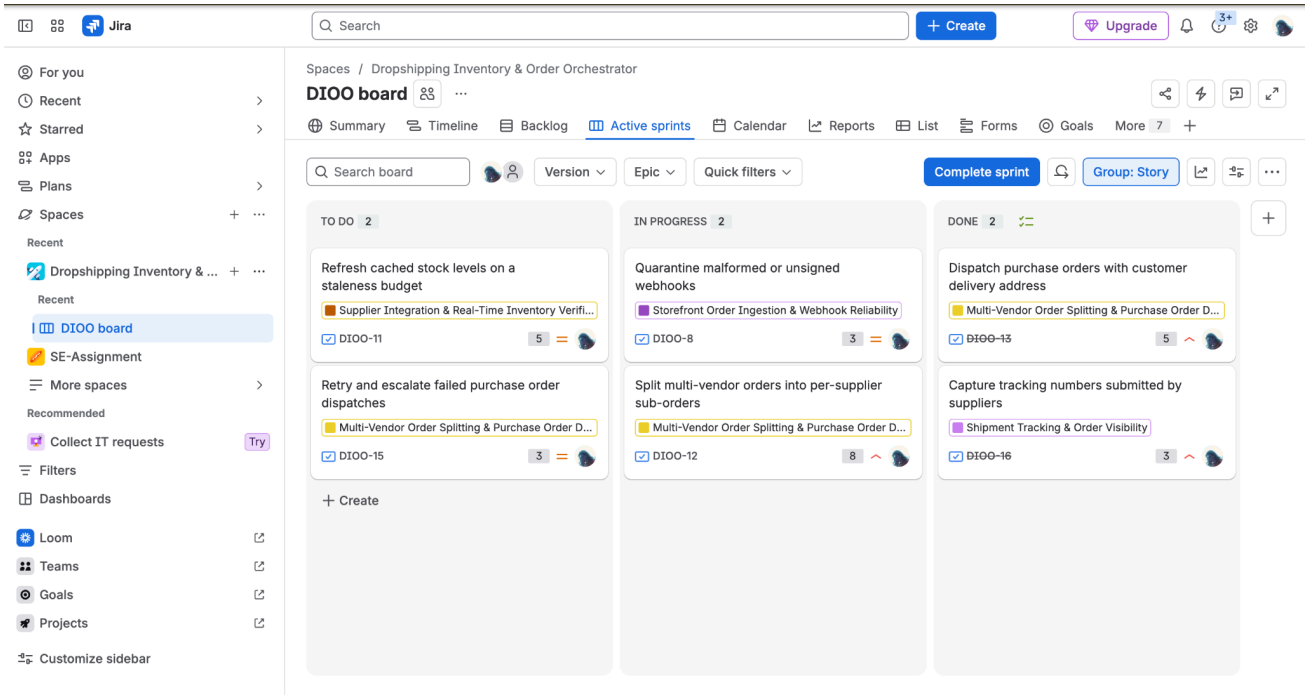
1.**Backlog**



2.**Story points**



3.Sprint board



4.Burndown

Jira

Search

Create

Upgrade

For you
Recent
Starred
Apps
Plans
Spaces
Recent
Dropshipping Inventory & ...
Recent
DIOO board
SE-Assignment
More spaces
Recommended
Collect IT requests Try
Filters
Dashboards
Loom
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Goals
Projects
Customize sidebar

Spaces / Dropshipping Inventory & Order Orchestrator / DIOO board / Reports

Burndown Chart

How to read this chart

How to read this chart

Track the total work remaining and project the likelihood of achieving the sprint goal. This helps your team manage its progress and respond accordingly.

DIOO Sprint 1 Story Points

Reliable webhook ingestion and live supplier stock verification

